

Subject-Aware EEG Denoising via Support-Calibrated Artifact-Subspace Diffusion

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Ocular propagation into EEG varies across users and recording contexts, while most learned denoisers apply a fixed population correction. We study query-disjoint calibration for unseen users through support-calibrated artifact-subspace diffusion. Support EEG/EOG determines a rank-two transfer basis by ridge regression, thin SVD, and within-montage Procrustes alignment. The basis structurally defines the diffusion coordinates, observation at every reverse step, and final EEG correction; participant identity and query EOG, labels, or outcomes are never model inputs. Diffusion predicts bounded artifact coefficients and is compared with an information-matched one-step estimator trained on the identical target. On 16 paired real-EEG/EOG-backed Klados source records, diffusion improves clean-RRMSE utility over the one-step estimator by 0.0592 (95% paired-bootstrap interval [0.0336, 0.0879]) and provides better uncertainty-error ranking. On 58 compatible SGEYESUB participant stems, the method satisfies frozen natural-EEG preservation margins. However, matching support does not improve the independent EOG endpoint over a population operator (-0.00515 , $[-0.0113, 0.00071]$) or a same-cell wrong operator (0.00286 , $[-0.00303, 0.00867]$). Thus iterative artifact-subspace diffusion has development-stage value under the tested target, while a specifically subject-aware calibration increment is not established.

CCS Concepts: • **Computing methodologies** → **Neural networks**; *Machine learning approaches*; • **Applied computing** → *Health informatics*.

Additional Key Words and Phrases: EEG denoising, denoising diffusion probabilistic model, subject-aware learning, cross-subject generalization, conditional generative model, brain-computer interface

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1 Introduction

Ocular artifacts are difficult to remove from electroencephalography (EEG) because their spatial propagation changes with montage, reference, electrode contact, and user physiology. A population denoiser can learn common artifact statistics, but a target user may have a different EOG-to-EEG transfer. The central question in this paper is deliberately narrow: can a short, query-disjoint calibration recording define an unseen user's corruption coordinates,

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and does iterative diffusion add value over a one-step estimator given exactly the same coordinates and targets?

We follow the principle *personalize the corruption, not the brain*. The clean-EEG model is never conditioned on participant identity. Instead, a rank-two artifact basis is estimated from support EEG/EOG by ridge regression, thin SVD, and within-montage Procrustes alignment. This basis has a structural role: it defines the low-dimensional diffusion state, the observation supplied at every reverse step, and the final EEG correction. Query EOG, eye tracking, artifact labels, outcomes, and participant identifiers are excluded from inference.

The resulting support-calibrated artifact-subspace diffusion model predicts bounded ocular coefficients rather than regenerating the full EEG. We compare it with an information-matched deterministic network trained on the identical target and with population and same-cell wrong-operator interventions using a shared checkpoint and common random numbers. This design separates two claims: (i) an incremental value of iterative diffusion, and (ii) an incremental value of matching-user calibration.

Experiments use paired real-EEG/EOG-backed semi-simulation for waveform fidelity and natural SGEYESUB recordings for participant-level artifact and preservation proxies. The evidence is intentionally reported as development evidence. The current experiment finds a clear advantage of the diffusion point estimate over the matched one-step estimator on the paired task, together with useful posterior risk ranking and acceptable natural-EEG safety. However, matching support does not reliably improve over population or same-cell wrong support. Thus the current data support artifact-subspace diffusion, but do not establish the stronger subject-aware claim.

Our contributions are:

- a query-disjoint corruption calibration whose basis directly defines diffusion coordinates and reconstruction, without participant embeddings;
- an information-matched deterministic comparison and operator interventions that isolate diffusion utility from subject-calibration value;
- participant/source-record paired evaluation with three training seeds, posterior uncertainty analysis, and explicit natural-EEG preservation tests;
- a scoped negative finding: under the tested support estimator, the matching-user increment is not identified even though iterative diffusion improves over the one-step estimator.

2 Related Work

2.1 EEG Artifact Correction

Classical ocular-artifact correction includes regression, filtering, and blind source separation. ICA is effective when artifact components are separable, but component selection and decomposition stability can vary with montage and recording duration. Regression methods use EOG references more directly, yet their transfer coefficients can differ strongly across participants and sessions. Deep denoisers trained on paired or semi-simulated examples avoid manual component labeling, including the EEGdenoiseNet line of work [2], but usually apply one population mapping at deployment.

Our work retains the interpretable EOG-to-EEG transfer while using a generative model only for low-dimensional artifact coefficients. The support recording is query-disjoint, and query EOG is reserved for evaluation. This differs from methods that require an EOG channel throughout deployment.

2.2 Cross-Participant Adaptation

EEG varies across participants because of anatomy, impedance, reference, behavior, and acquisition state. Many subject-aware networks encode a training participant label or learn a closed-set identity embedding. Such an embedding cannot directly represent a previously unseen user and risks personalizing neural content rather than the acquisition corruption. Domain-adaptation methods address population shift but typically do not identify the causal role of a short support calibration.

We instead personalize only the pollution operator. A target participant is represented by a support-derived artifact basis, not by an ID. Matching, population, and same-cell wrong interventions share network weights and queries, which makes the subject-calibration increment directly testable.

2.3 Diffusion for Biosignal Denoising

Diffusion models offer stable denoising objectives and posterior samples without an adversarial discriminator [1]. Existing biosignal applications include ECG reconstruction [3] and noisy-signal-conditioned EEG models. Conditioning may describe the noisy observation, an artifact type, a domain representation, or a participant label. These alternatives do not by themselves ensure that a model uses an unseen user's calibration; a sufficiently strong network may ignore a soft context vector.

The present construction makes support information structural. The calibrated basis defines the diffusion coordinate system, the observation condition at every reverse step, and the reconstruction operator. It also differs from full-EEG diffusion: only two bounded artifact-coefficient channels are diffused, and the observed orthogonal complement is copied exactly. Finally, we compare against a capacity-matched one-step model trained on the identical coefficient target, preventing a full-generation-versus-regression mismatch from being misinterpreted as diffusion utility.

3 Support-Calibrated Artifact-Subspace Diffusion

3.1 Problem and Information Boundary

Let $\mathbf{y} \in \mathbb{R}^{C \times L}$ be an observed query EEG segment. For an unseen participant or context s , a disjoint calibration set contains only synchronized support EEG and EOG, $\mathcal{S}_s = \{\mathbf{Y}_s^{\text{cal}}, \mathbf{E}_s^{\text{cal}}\}$. No query EOG, eye tracking, artifact label, outcome, clean target, or participant identifier is visible to a denoising method. We model ocular contamination as

$$\mathbf{y} = \mathbf{x} + \mathbf{A}_s \mathbf{a}_s + \boldsymbol{\epsilon}, \quad (1)$$

where $\mathbf{A}_s \in \mathbb{R}^{C \times r}$ is support-derived and $r = 2$. The model personalizes only this corruption geometry; it does not learn a participant-specific clean-EEG prior.

3.2 Support-Only Artifact Geometry

Support EEG and EOG are centered and standardized within each exact montage-reference-sampling-rate cell. We estimate the transfer in FP64 as

$$\hat{\mathbf{C}}_s = \mathbf{Y}_s^{\text{cal}} (\mathbf{E}_s^{\text{cal}})^{\top} (\mathbf{E}_s^{\text{cal}} (\mathbf{E}_s^{\text{cal}})^{\top} + \lambda \mathbf{I})^{-1}, \quad (2)$$

implemented with a linear solve. A thin SVD of $\hat{\mathbf{C}}_s = \mathbf{U}_s \boldsymbol{\Sigma}_s \mathbf{V}_s^{\top}$ provides its retained left subspace. Effective-rank deficiencies are retained in a binary rank mask rather than hidden by rank search.

SVD coordinates are ambiguous up to sign and within-subspace rotation. For each compatible cell, a population basis $\mathbf{U}_0$ is fitted using only outer-training participants. The support basis is aligned by orthogonal Procrustes,

$$\mathbf{A}_s = \mathbf{U}_s \mathbf{R}_s, \quad \mathbf{R}_s = \arg \min_{\mathbf{R}^\top \mathbf{R} = \mathbf{I}} \|\mathbf{U}_s \mathbf{R} - \mathbf{U}_0\|_F. \quad (3)$$

No alignment crosses incompatible cells. Support reliability $\rho_s \in [0, 1]$ depends only on effective rank, transfer stability, and support cross-validation error.

3.3 Bounded Artifact-Coefficient Target

On paired training examples, the physical coefficient target is

$$\mathbf{a}_s^* = \mathbf{A}_s^\top (\mathbf{y} - \mathbf{x}_{\text{clean}}). \quad (4)$$

A robust, training-only cell scale τ defines the same bounded target for diffusion and deterministic models:

$$\mathbf{u}_s^* = \tanh(\mathbf{a}_s^* / \tau), \quad \mathbf{u}_s^* \in [-1, 1]^{r \times L}. \quad (5)$$

Training batches mix artifact-active examples and training-only low-EOG identity examples in a fixed 1:1 ratio; identity examples use $\mathbf{u}^* = 0$. This teaches the cleaner not to alter low-artifact EEG.

3.4 Coefficient Diffusion

The diffusion state has only $r = 2$ channels. Under a cosine schedule,

$$\mathbf{u}_t = \sqrt{\alpha_t} \mathbf{u}_s^* + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}. \quad (6)$$

A 1-D U-Net predicts the v parameterization. At every reverse step its inputs include $\mathbf{u}_t$, the timestep, the coordinate observation $\mathbf{A}_s^\top \mathbf{y}$, ρ_s , the rank mask, and valid-time mask. Reliability modulates hidden blocks through FiLM. We use EMA weights, Min-SNR weighting, one frozen DDIM sampler with 25 network evaluations, and $K = 8$ fixed random streams. The posterior point estimate is the arithmetic mean, never a target-selected sample:

$$\bar{\mathbf{u}}_s = K^{-1} \sum_{k=1}^K \hat{\mathbf{u}}_s^{(k)}. \quad (7)$$

The physical correction and output are

$$\Delta_s = \mathbf{A}_s \tau \bar{\mathbf{u}}_s, \quad \hat{\mathbf{x}}_s = \mathbf{y} - \Delta_s. \quad (8)$$

Consequently, $(\mathbf{I} - \mathbf{A}_s \mathbf{A}_s^\top) \hat{\mathbf{x}}_s = (\mathbf{I} - \mathbf{A}_s \mathbf{A}_s^\top) \mathbf{y}$ up to numerical tolerance. This guarantees preservation of the observed component orthogonal to the *estimated artifact basis*; it is not a claim that all neural signal is preserved.

3.5 Population Fallback and Operator Interventions

The safe correction mixes matching and population branches:

$$\Delta_{\text{final}} = \rho_s \Delta_s + (1 - \rho_s) \Delta_0. \quad (9)$$

When $\rho_s = 0$, execution short-circuits before constructing or sampling the matching branch. Subject awareness is tested by replacing only the operator: DIFF-MATCH uses $\mathbf{A}_s$, DIFF-POP uses $\mathbf{A}_0$, and DIFF-WRONG uses a different participant's operator from the same compatibility cell. All three share weights, queries, normalization, reliability, and common random numbers.

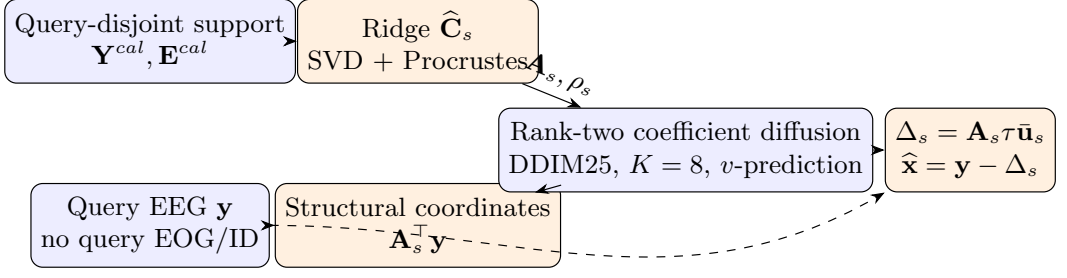


Fig. 1. Support-calibrated artifact-subspace diffusion. The support-derived basis is not merely a soft feature: it defines the query coordinates at every reverse step and maps predicted coefficients back to the EEG correction.

3.6 Information-Matched Deterministic Comparator

DET-MATCH predicts the identical target in one pass,

$$\hat{\mathbf{u}}_{\text{det}} = g_\phi(\mathbf{A}_s^\top \mathbf{y}, \rho_s, \mathbf{m}_s), \quad \hat{\mathbf{x}}_{\text{det}} = \mathbf{y} - \mathbf{A}_s \tau \hat{\mathbf{u}}_{\text{det}}. \quad (10)$$

It shares participants, targets, identity examples, normalization, windowing, maximum updates, and backbone scale with diffusion. Each method selects its own validation-optimal checkpoint. Three fixed training seeds are averaged within each statistical unit before inference.

4 Experiments

4.1 Questions and Frozen Comparisons

We ask whether iterative diffusion improves over a one-step estimator (H1), whether matching support improves over population and same-cell wrong support (H2), and whether natural-EEG preservation remains acceptable (H3). The core methods are RAW, an existing population deterministic cleaner (POP), DIFF-POP, DIFF-MATCH, DIFF-WRONG, and DET-MATCH. The three diffusion arms use the same checkpoint and common random numbers. We additionally report $K = 1$ versus $K = 8$ and removal of identity examples as targeted ablations.

All models use three fixed training seeds (20260811–20260813), at most 12,000 updates, independent validation-optimal checkpoints, and the same window and normalization pipeline. Diffusion uses a cosine schedule, v -prediction, EMA, Min-SNR weighting, DDIM25, and an arithmetic mean of eight samples. Statistical units are source records or participant stems, never windows. Windows and seeds are first averaged within a unit; 20,000 paired bootstrap samples then provide descriptive 95% intervals. SGE bootstrap sampling is stratified by exact montage–reference–sampling-rate cell.

4.2 Datasets and Evidence Scope

Klados–Bamidis v4. We use the frozen split with source records 1–30 for training, 31–36 and 44–45 for development, and all 16 records 37–43 and 46–54 for evaluation. The recordings combine real EEG and real ocular sources with a common clean target. Participant provenance is not reliable, so this is paired source-record mechanism evidence, not participant-held-out

Table 1. Klados paired mechanism results (three-seed averages).

Method	Clean RRMSE ↓	Correlation ↑	ΔSNR dB ↑
RAW	1.2948	0.6302	0.000
POP	0.3990	0.9204	10.402
DIFF-POP	0.4767	0.8932	8.925
DIFF-MATCH	0.5325	0.8744	7.940
DIFF-WRONG	0.5543	0.8641	7.751
DET-MATCH	0.5917	0.8497	7.017

evidence. The primary endpoint is clean RRMSE; correlation, artifact reconstruction error, ΔSNR, and complement error are secondary.

SGEYESUB. All five studies are used as real-EEG development data. For each participant stem, block 1 is support and block 2 is query. Operators and wrong controls never cross exact compatibility cells. Query EOG and annotations are opened only after all method outputs are frozen. Of 59 available stems, 58 are compatible; study05/study05_p42 lacks a compatible population cell and remains in the coverage denominator. The primary artifact endpoint is independent EOG coherence reduction. Safety endpoints are low-artifact observation preservation, ERP preservation, continuous-segment Welch PSD distortion, reference-free covariance distortion, output scale, and observation change. No clean-waveform claim is made for SGEYESUB.

Independent participant data. The official EEGEyeNet OSF page redirects its synchronized release to a public Google Drive. Two scheduled, resumable attempts reached the participant MAT folders but the official server denied public access to participant files. Consequently, no independent EEGEyeNet outcome was opened and the current experiment remains development evidence.

4.3 Technical Validity

The implementation passed bounded-target overfit, zero-correction identity, finite reverse trajectory, coefficient bounds, complement consistency ($\leq 10^{-5}$), operator intervention, EMA checkpoint reload/resume, explicit CUDA generator state, participant-specific seeds, and forbidden-query-field tests. Training completed all 78 dataset–fold–seed tasks. A first Klados evaluation attempt exposed an FP64/FP32 mismatch in the diagnostic complement metric; only that post-processing conversion was corrected, and all three Klados inference seeds were replayed.

4.4 Paired Waveform Results (H1)

Table 1 reports source-record means. Lower RRMSE is better. DIFF-MATCH improves over the information-matched DET-MATCH by a utility effect of 0.0592 (95% CI [0.0336, 0.0879], $n = 16$), supporting H1 under the tested protocol. POP remains stronger than either learned matching model; H1 therefore identifies an iterative-versus-one-step difference, not superiority over every baseline.

The eight-sample posterior also supplies useful uncertainty. Its uncertainty–error correlation is 0.445 versus 0.140 for a three-seed deterministic ensemble, and its risk–coverage AUC is

Table 2. SGEYESUB real-EEG development results. EOG reduction and preservation are higher-is-better; PSD distortion is lower-is-better.

Method	EOG reduction	Preservation	PSD distortion
RAW	0.0000	1.0000	0.0000
POP	0.3231	0.7762	0.3830
DIFF-POP	0.2375	0.8973	0.1509
DIFF-MATCH	0.2323	0.8891	0.1602
DIFF-WRONG	0.2295	0.8938	0.1543
DET-MATCH	0.2301	0.8683	0.2039

lower (0.241 versus 0.292). These are development diagnostics rather than an independently confirmed clinical risk model.

4.5 Natural EEG Subject Intervention and Safety (H2–H3)

Table 2 shows natural-EEG means. Matching support does not improve the independent EOG endpoint over population: $\Delta_{\text{match-pop}} = -0.00515$ ($[-0.0113, 0.00071]$, $n = 58$). Its advantage over a same-cell wrong operator is 0.00286 ($[-0.00303, 0.00867]$), also inconclusive. H2 is therefore not supported.

Against the frozen -0.02 population-relative margin, DIFF-MATCH passes all four safety endpoints: low-artifact preservation, ERP preservation, continuous-segment Welch PSD distortion, and covariance distortion. Identity examples materially improve safety: removing them reduces preservation from 0.889 to 0.796 and increases PSD distortion from 0.160 to 0.358. H3 is supported in development.

4.6 Interpretation

The complete result is mixed and constrains the paper’s claim. Iterative coefficient diffusion improves over an information-matched one-step model and its posterior variance provides better error ranking. The structural operator conditioning is numerically active and safe, but the matching support operator does not outperform population or same-cell wrong support at the participant level. Thus the current experiment supports diffusion in an estimated artifact subspace, while the stronger subject-aware transfer claim remains unverified.

5 Conclusion

We introduced a support-calibrated artifact-subspace diffusion model that personalizes an estimated ocular propagation geometry rather than a clean-brain prior or participant identity. The support basis defines the rank-two latent state and the observation-preserving reconstruction, making it impossible for the implementation to ignore the operator while still producing the declared output.

On 16 paired Klados source records, the diffusion point estimate improved clean RRMSE utility over an information-matched deterministic estimator by 0.0592 (95% paired-bootstrap interval $[0.0336, 0.0879]$). Its sample variance also ranked errors more effectively than a three-seed deterministic ensemble. On 58 of 59 SGEYESUB stems, the frozen natural-EEG preservation margins passed. In contrast, matching calibration did not improve the independent EOG endpoint over the population operator (-0.00515 , $[-0.0113, 0.00071]$) or the same-cell wrong operator (0.00286, $[-0.00303, 0.00867]$). The supported conclusion is

therefore limited: iterative diffusion is useful for the tested artifact-subspace target, but a subject-specific increment from the present support estimator is not established.

Klados records are not reliable participant identities and SGEYESUB was used in development. Moreover, the public EEGEyeNet release could not be retrieved because its official Google Drive denied repeated participant-file access. Independent participant-level confirmation remains necessary before presenting the method as a validated subject-aware denoiser.

Ethics and Privacy Statement

This study uses de-identified research EEG released by the original dataset providers; no new human-subject data were collected. The method does not learn participant identifiers or biometric embeddings. A target user's short support recording is used only to estimate an artifact propagation basis, and query EOG, eye tracking, labels, and outcomes are excluded from inference. EEG remains sensitive physiological data and should be handled under the access and privacy terms of each source dataset.

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A Subspace Reconstruction Identity

Let $\mathbf{A} \in \mathbb{R}^{C \times r}$ have orthonormal columns and let $\hat{\mathbf{x}} = \mathbf{y} - \mathbf{A}\mathbf{d}$ for any coefficient sequence $\mathbf{d} \in \mathbb{R}^{r \times L}$. Defining $\mathbf{Q} = \mathbf{I} - \mathbf{A}\mathbf{A}^\top$,

$$\mathbf{Q}\hat{\mathbf{x}} = \mathbf{Q}\mathbf{y} - \mathbf{Q}\mathbf{A}\mathbf{d} = \mathbf{Q}\mathbf{y},$$

because $\mathbf{Q}\mathbf{A} = \mathbf{A} - \mathbf{A}(\mathbf{A}^\top\mathbf{A}) = \mathbf{0}$. The implementation evaluates the relative residual of this identity on valid time samples and requires it to be at most 10^{-5} . The identity says only that the observation is preserved outside the *estimated* artifact basis.

B Bounded Coefficients

For a positive training-only scale τ_j and physical coefficient a_j , the target $u_j = \tanh(a_j/\tau_j)$ lies in $(-1, 1)$. The sampler applies the same bound to every predicted coefficient before reconstruction. Therefore the per-coordinate physical correction satisfies $|\tau_j u_j| \leq \tau_j$. This bound is independent of the evaluation target and prevents a reverse trajectory from producing an unbounded EEG correction.

C Reporting Scope

Klados evaluation units are source records, not participants. SGEYESUB units are participant stems, but all SGE studies were available during development. The intervals in the main paper are paired-bootstrap development intervals and must not be interpreted as independent confirmation. The attempted independent EEGEyeNet evaluation was blocked before test outcomes were opened.

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